



Project-05

**Bachelor of Computer Application – AI ML
Semester – V**

Sub: Deep Learning

Topic: Activation Functions Using Python

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Activation Function Using Python

```
import numpy as np
import matplotlib.pyplot as plt

#Task 1 : define activation function

#sigmoid functions def
sigmoid(x):    return 1/(1+
np.exp(-x))

#relu function def
relu(x):    return
np.maximum(0,x)

# Tanh Function def
tanh(x):    return
np.tanh(x)

#Task 2

#input values
x_value = np.array([-5,-3,-1,0,1,3,5])

#Apply activation finctions
sigmoid_output = sigmoid(x_value)
relu_output = relu(x_value)
tanh_output = tanh(x_value)

#display outputs
print("input Values")
print(x_value)

print("\nSigmoid Output")
print(sigmoid_output)
input Values
[-5 -3 -1  0  1  3  5]
Sigmoid Output
[0.00669285 0.04742587 0.26894142 0.5          0.73105858 0.95257413
 0.99330715]

#Task 3
#generated continous input values for plotting
```

```
x= np.linspace(-5,5,100)
```

```
#calculate outputs y_sigmoid = sigmoid(x) y_relu = relu(x) y_tanh =  
tanh(x)
```

```
#plot all activation functions plt.figure(figsize=(10,6))  
plt.plot(x,y_sigmoid,label='sigmoid') plt.plot(x,y_relu,label='relu') plt.plot(x,y
```

```
plt.title("Comparison Of Activation Finctions") plt.xlabel("input value") plt.ylabel("output Value")
```

```
plt.grid(True) plt.legend() plt.show()
```

